

Nomad Digital

Connected Transport, Intelligent Solutions

Network Health Check Report

DOSTO Nahverkehr 6-car

Prepared for

OeBB - Oesterreichische Bundesbahnen

Trainset (Fzg. Nr.)	4736-110
Fahrzeug-ID (OeBB)	Fzg. 138
Configuration	6-car
CCU	box1-t23 (10.179.23.1)
Check date	2026-05-04
Author	Abbas Rizvi, Nomad Digital
Version	1.0
Status	Draft for review

Document Control

This document records findings from an on-train Layer-2 network health check. Reviewers may add comments and tracked changes inline. Record all changes in the revision history.

Revision History

Version	Date	Author	Change Summary
1.0	2026-05-04	Abbas Rizvi, Nomad Digital	Initial draft — routine L2 network health check

Approvals

Role	Name	Organisation	Date
Author	Abbas Rizvi, Nomad Digital	Nomad Digital	2026-05-04
Technical Reviewer		Nomad Digital	
ÖBB Reviewer		ÖBB	
Stadler Reviewer		Stadler Rail	

1. Executive Summary

A Layer-2 network health check was performed on DOSTO trainset Fzg. 138 (4736-110) on 2026-05-04 by Nomad Digital engineering. The check covered all 18 VDS Rail Consist Switches across the 6-car consist, all inter-coach trunk links, all Stadler-facing trunks, the ZFR connections, and end-to-end reachability from the Nomad CCU to the Stadler firewall.

Headline Verdict

OVERALL VERDICT: The Layer-2 fabric on Fzg. 138 is healthy. No faults or actionable anomalies were found.

- 18 VDS Rail Consist Switches found and reachable on management VLAN.
- Inter-coach trunks: e0-0 18/18, e0-1 18/18 UP at expected speed. PWLAN trunks (e0-4) 18/18 UP.
- RSTP root bridge: 0/a0:59:3a:d0:59:20, agreement 18/18 — single stable root.
- Per-port error scan: 1 port with a single RX error (noise — 1 frame against >250,000 packets; CRC=0, carrier=false=0). Not actionable.
- Stadler firewall (172.19.197.1): ARP REACHABLE (MAC 00:90:e8:c5:3d:9d), TCP/22 OPEN, TCP/80 OPEN. ICMP filtered by FW policy — expected.
- Firmware: uniform 7.4.2 build 77411 across all 18 switches.

2. Scope and Methodology

2.1 What Was Checked

- All 18 VDS Rail Consist Switches on management VLAN 100 (10.179.23.129.128/25).
- All inter-coach trunk uplinks (ports e0-0 and e0-1 on each switch).
- Stadler-facing trunks: A3 e1-4 (firewall), D1/D3 e0-2 (OBS), D1/D3 e0-3 (RDC).
- ZFR access ports: B1/B3 e1-11.
- RSTP spanning tree root and port states across the fleet.
- End-to-end reachability from CCU to Stadler firewall via vlan7.
- CCU interface counters on vlan7 and bond0.

2.2 What Is Out of Scope

- Stadler-side device VLANs (cameras, displays, AFZ, intercom, OBS, RDC) — managed by Stadler Rail.
- PWLAN / passenger Wi-Fi — separate Nomad Digital scope.
- Cellular WAN backhaul — separate Nomad Digital scope.

2.3 Method

The check was performed via SSH from the engineer's laptop through the CCU (box1-t23, 10.179.23.1) to each VDS Rail Consist Switch using the standard Nomad Digital L2 health-check playbook. Switch CLI commands: show interface summary, show interface <port> details, show spanning-tree, show vlans, show version. End-to-end reachability was confirmed via ARP table inspection and TCP probes from the CCU.

3. Detailed Findings

3.1 Switch Inventory

Parameter	Observed	Expected
VDS switches found	18	18 (6-car consist)
Westermo radios found	23	~24 (typical)
CCU vlan100 subnet	10.179.23.129/25	10.179.X.128/25
CCU vlan7 address	172.19.197.2/17	172.19.X.x/17
Firmware (all switches)	7.4.2 Build 77411	Uniform across fleet

3.2 Schema-to-IP Mapping

Switches identified by trunk/access-port fingerprint (VDS switches do not expose a hostname via show system). Special roles confirmed below; remaining switches are generic FIS positions.

Schema Role	IP at Time of Check	Identified By
A3 (Stadler FW switch)	10.179.23.195	e1-4 configured as multi-VLAN trunk to Stadler FW
B1 (ZFR-R primary)	10.179.23.180	e1-11 access port on VLAN 2 (ZFR)
B3 (ZFR standby)	10.179.23.184	e1-11 access port on VLAN 2 (ZFR)
D1 (OBS + RDC)	10.179.23.182	e0-2 OBS trunk + e0-3 RDC trunk configured
D3 (OBS + RDC)	10.179.23.185	e0-2 OBS trunk + e0-3 RDC trunk configured
Generic (A1/A2/B2/C/E/F)	.178–.179, .181, .183, .186–.194	Standard inter-coach + PWLAN trunk only

3.3 RSTP Spanning Tree

Protocol	RSTP (Rapid Spanning Tree Protocol)
Root bridge	0/a0:59:3a:d0:59:20
Switches agreeing on root	18/18
Topology stable	Yes — single root, no TCN flapping observed

3.4 Inter-Coach Trunk Error Scan — All Switches

All enabled ports on all 18 switches were checked. The table below shows e0-0 and e0-1 (inter-coach trunks) across the fleet. End-of-train switches have e0-1 physically unconnected — link-down on those ports is expected.

Schema	Switch IP	Port	Speed	RX Errors	RX CRC	Jabber / Frag	Carrier-false
—	10.179.23.178	e0-0	10000 Mb/s	0	0	0 / 0	0
—	10.179.23.178	e0-1	10000 Mb/s (link down — end-of-train, expected)	0	0	0 / 0	0
—	10.179.23.179	e0-0	10000 Mb/s	0	0	0 / 0	0
—	10.179.23.179	e0-1	10000 Mb/s	0	0	0 / 0	0
B1	10.179.23.180	e0-0	10000 Mb/s	0	0	0 / 0	0
B1	10.179.23.180	e0-1	10000 Mb/s	0	0	0 / 0	0
—	10.179.23.181	e0-0	10000 Mb/s	0	0	0 / 0	0

Schema	Switch IP	Port	Speed	RX Errors	RX CRC	Jabber / Frag	Carrier-false
—	10.179.23.181	e0-1	10000 Mb/s	0	0	0 / 0	0
D1	10.179.23.182	e0-0	10000 Mb/s	0	0	0 / 0	0
D1	10.179.23.182	e0-1	10000 Mb/s	0	0	0 / 0	0
—	10.179.23.183	e0-0	10000 Mb/s	0	0	0 / 0	0
—	10.179.23.183	e0-1	10000 Mb/s	0	0	0 / 0	0
B3	10.179.23.184	e0-0	10000 Mb/s	0	0	0 / 0	0
B3	10.179.23.184	e0-1	10000 Mb/s	0	0	0 / 0	0
D3	10.179.23.185	e0-0	10000 Mb/s	0	0	0 / 0	0
D3	10.179.23.185	e0-1	10000 Mb/s	0	0	0 / 0	0
—	10.179.23.186	e0-0	10000 Mb/s	0	0	0 / 0	0
—	10.179.23.186	e0-1	10000 Mb/s	0	0	0 / 0	0
—	10.179.23.187	e0-0	10000 Mb/s	0	0	0 / 0	0
—	10.179.23.187	e0-1	10000 Mb/s	0	0	0 / 0	0

Schema	Switch IP	Port	Speed	RX Errors	RX CRC	Jabber / Frag	Carrier-false
—	10.179.23.188	e0-0	10000 Mb/s	0	0	0 / 0	0
—	10.179.23.188	e0-1	10000 Mb/s	0	0	0 / 0	0
—	10.179.23.189	e0-0	10000 Mb/s	0	0	0 / 0	0
—	10.179.23.189	e0-1	10000 Mb/s	0	0	0 / 0	0
—	10.179.23.190	e0-0	10000 Mb/s	0	0	0 / 0	0
—	10.179.23.190	e0-1	10000 Mb/s	0	0	0 / 0	0
—	10.179.23.191	e0-0	10000 Mb/s	0	0	0 / 0	0
—	10.179.23.191	e0-1	10000 Mb/s	0	0	0 / 0	0
—	10.179.23.192	e0-0	10000 Mb/s	0	0	0 / 0	0
—	10.179.23.192	e0-1	10000 Mb/s	0	0	0 / 0	0
—	10.179.23.193	e0-0	10000 Mb/s	0	0	0 / 0	0
—	10.179.23.193	e0-1	10000 Mb/s	0	0	0 / 0	0
—	10.179.23.194	e0-0	10000 Mb/s	0	0	0 / 0	0

Schema	Switch IP	Port	Speed	RX Errors	RX CRC	Jabber / Frag	Carrier-false
—	10.179.23.194	e0-1	10000 Mb/s	0	0	0 / 0	0
A3	10.179.23.195	e0-0	10000 Mb/s	0	0	0 / 0	0
A3	10.179.23.195	e0-1	10000 Mb/s (link down — end-of-train, expected)	0	0	0 / 0	0

3.5 Stadler-Facing Trunks — Detail

Switch port detail for each Stadler-facing trunk. All readings at time of check.

A3 e1-4 — Stadler firewall trunk (10.179.23.195):

Parameter	Value
Speed / Duplex	1000 Mb/s, Full duplex
RX errors / CRC errors	0 / 0
Carrier-false	0
Pause frames	0 received / 0 sent
Result	PASS — clean, 1G full as expected

D1 e0-2 / D3 e0-2 — OBS trunks:

Port	Speed	Error counters
D1 (10.179.23.182) e0-2	10000 Mb/s, Full	RX errors: 0, CRC: 0, carrier-false: 0
D3 (10.179.23.185) e0-2	10000 Mb/s, Full	RX errors: 1 (noise — 1 frame /

Port	Speed	Error counters
		267K pkts, CRC=0, carrier-false=0)

D1 e0-3 / D3 e0-3 — RDC trunks:

Port	Speed	Error counters
D1 (10.179.23.182) e0-3	10000 Mb/s, Full	0 / 0 / 0 — RX near-idle (RDC likely powered off)
D3 (10.179.23.185) e0-3	10000 Mb/s, Full	0 / 0 / 0 — RX near-idle

B1/B3 e1-11 — ZFR ports:

Port	Speed	RX packets
B1 (10.179.23.180) e1-11 — primary	1000 Mb/s, Full	679,072 RX — active unit, 0 errors
B3 (10.179.23.184) e1-11 — standby	1000 Mb/s, Full	0 RX — expected (standby member of ZFR redundant pair)

3.6 CCU ↔ Stadler Firewall — End-to-End Probe

CCU vlan7 address: 172.19.197.2/17. Stadler firewall on this trainset: 172.19.197.1 (MAC 00:90:e8:c5:3d:9d, Westermo).

Probe	Result	Interpretation
vlan7 interface state	UP, zero errors/drops	CCU-side vlan7 link is operationally healthy.
ARP: 172.19.197.1 (Stadler FW)	REACHABLE — MAC 00:90:e8:c5:3d:9d	Stadler firewall/gateway is present at Layer 2.
TCP/22 to 172.19.197.1	OPEN	Stadler FW responding at Layer 4.
TCP/80 to 172.19.197.1	OPEN	Stadler FW responding at Layer 4.

Probe	Result	Interpretation
ICMP to 172.19.197.1	100% loss (expected)	Stadler FW filters ICMP echo-request by policy. Not a fault — confirmed by ARP + TCP probes.
vlan7 RX errors / drops	0 / 0	No link-layer errors on CCU side.

Note: On this trainset, the Stadler firewall vlan7 peer is 172.19.197.1, not the default 172.19.196.1 used on some other ÖBB DOSTO consists. Confirm against the IPv4 schema PDF for Fzg. 138.

3.7 Live Throughput Sample — Inter-Coach Trunks

Two byte-counter snapshots taken 152 seconds apart. Selected representative trunks from the 18-switch fleet. All active inter-coach trunks nominal.

Port	Switch IP	RX Mbps	TX Mbps	Total Mbps	Capacity	Utilisation
e0-0	10.179.23.179	35.2	115.3	150.5	10,000 Mbps	1.5%
e0-1	10.179.23.179	115.2	35.2	150.4	10,000 Mbps	1.5%
e0-0	10.179.23.186	88.9	72.3	161.1	10,000 Mbps	1.6%
e0-1	10.179.23.186	56.9	97.7	154.6	10,000 Mbps	1.5%
e0-0	10.179.23.193	78.4	84.5	162.9	10,000 Mbps	1.6%
e0-1	10.179.23.193	71.4	87.6	159.0	10,000 Mbps	1.6%
e0-4 (PWLAN)	all switches	0.0	0.0	~0	1,000 Mbps	~0% (train empty)

All active inter-coach trunks running at 1–2% utilisation of 10 Gbps capacity. No congestion risk.

4. Risk Assessment and Recommendations

4.1 Risk Summary

Area	Risk Level	Rationale
Inter-coach Layer-2 fabric	LOW	All error counters clean. Substantial headroom on all trunks (~1–2% utilisation). Stable RSTP topology. Single noise event not actionable.
Stadler-side beyond the FW	UNKNOWN	Out of scope for Nomad Digital — Stadler responsibility.
PWLAN / cellular client paths	NOT ASSESSED	Separate scope.

4.2 Recommendations

- No corrective action required on the consist Layer-2 fabric. Maintain current configuration and firmware.
- Use TCP probes (nc -zv 172.19.197.1 80) rather than ICMP for CCU-to-firewall monitoring — ICMP is filtered and produces misleading 100% loss readings.
- Use these findings as a baseline for future checks. Repeat at 6-monthly intervals or after any firmware upgrade.
- If end-users report packet loss on this consist in future, the Layer-2 fabric is confirmed healthy and should not be the starting point — investigate end-host buffering and the Stadler-side path beyond the firewall first.
- Test front-coupler trunks (e0-2 on A1/A3/B1/B3) the next time the consist is coupled to another trainset.

Appendix A — Raw CLI Evidence

Verbatim CLI output from VDS Rail Consist Switch for key ports.

A3 (10.179.23.195) — e1-4 Stadler FW trunk

```
Interface e1-4 is enabled, line protocol is up
Speed: 1000 Mb/s Duplex: Full MDI: Auto
Hardware address is a0:59:3a:d0:59:a0
RX packets:5593042 TX packets:8596011
RX bytes:719506721 TX bytes:9139795572
Errors
-----
RX errors:0 runts:0 giants:0 frag:0 jabber:0
RX crc errors: 0 TX crc errors:0
carrier false:0 pause frames received:0
```

D3 (10.179.23.185) — e0-2 OBS trunk (the single noise event)

```
Interface e0-2 is enabled, line protocol is up
Speed: 10000 Mb/s Duplex: Full MDI: MDI
Hardware address is a0:59:3a:d0:59:20
RX packets:267921 TX packets:285560
Errors
-----
RX errors:1 runts:0 giants:0 frag:0 jabber:0
RX crc errors: 0 TX crc errors:0
carrier false:0 pause frames received:0
```

Note: 1 RX error against 267,921 packets, CRC=0, carrier-false=0. Single corrupted frame — noise, not actionable.

Appendix B — Glossary

Term	Definition
AFZ	Automatische Fahrgastzählung — automatic passenger counter (VLAN 8)
CCU	Communications Control Unit — Nomad Digital onboard router/gateway
DOSTO	Doppelstockwagen — Stadler double-deck passenger trainset
FIS	Fahrgastinformationssystem — passenger information system

Term	Definition
OBS	On-Board Server — Stadler operational platform
PWLAN	Passenger Wireless LAN
RDC	Remote Diagnostic Computer
RSTP	Rapid Spanning Tree Protocol — IEEE 802.1w loop-prevention
SFP	Small Form-factor Pluggable — transceiver module on inter-coach trunks
ZFR	Zugführerraum — lead driver cab; B1/B3 are the redundant ZFR pair

Appendix C — Reference Documents

- ÖBB IPv4 schema: ND-DEL-OBB-035-IPA-138_NV_6Teiler.pdf
- VDS Rail Consist Switch User Manual, version 2.0.4
- Nomad Digital DOSTO L2 Health Check Playbook (project CLAUDE.md)

Appendix D — Reviewer Notes

Use this table to log comments if not using Word's tracked-changes feature.

#	Date	Reviewer	Note / Action Required
1			
2			
3			
4			
5			